

Activity 2: Structure Density Reasoning

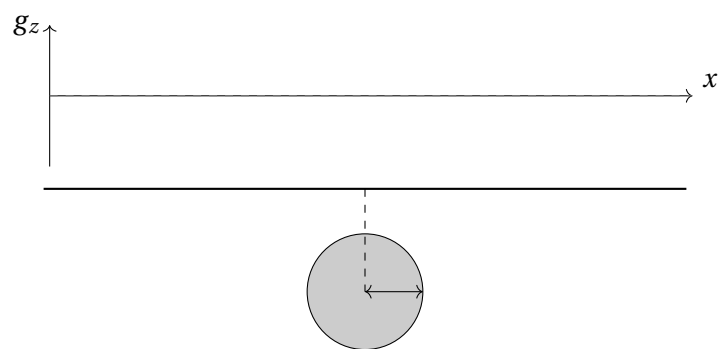
For each assigned structure, complete the table and use the *upper axes* to sketch the expected gravity anomaly $g_z(x)$ (arbitrary units). The *lower panel* shows a schematic cross-section (z positive downward).

Table

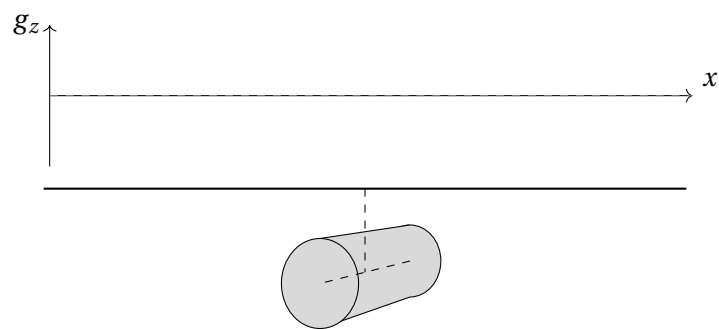
Structure	Likely lithologies	Plausible $\Delta\rho$ (kg m^{-3})
A.		
B.		
C.		
D.		
E.		
F.		
G.		

Prompts: Indicate sign (pos./neg.), width (narrow/broad), symmetry/asymmetry, and where the steepest gradients occur.

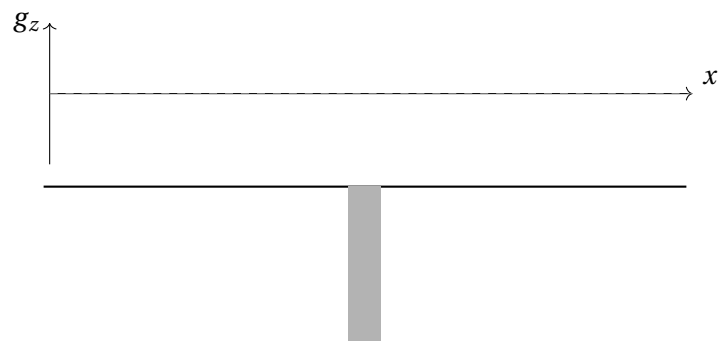
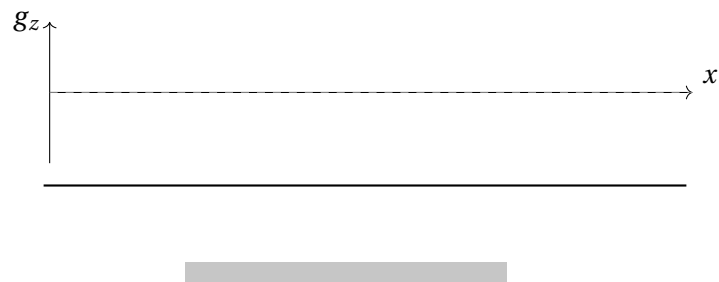
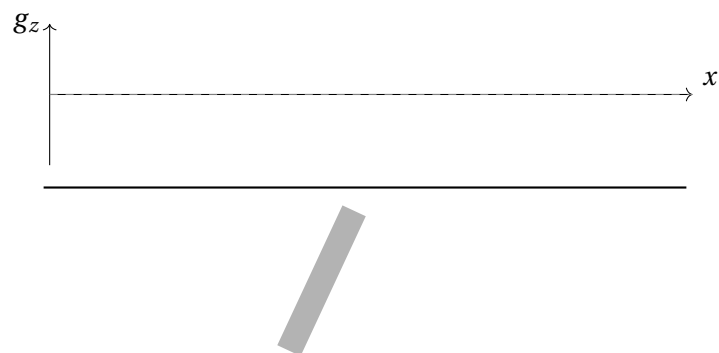
A) Buried Sphere (3D)

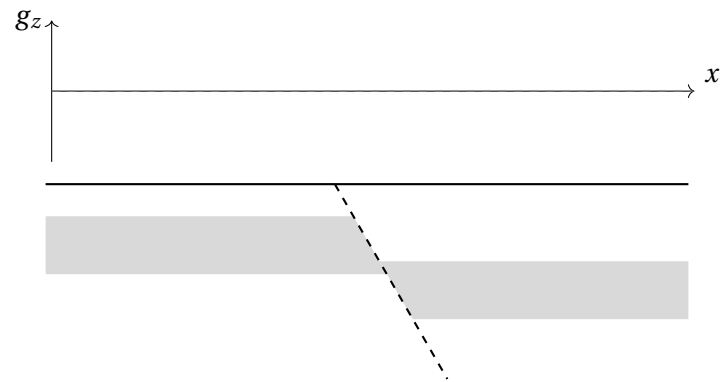


B) Buried Cylinder (2D; infinite strike)



C) Vertical Slab

**D) Finite-Extent Horizontal Slab****E) Inclined Plane**

F) Offset Layer**G) Gently Sloped Basin (edge at one-third from left boundary)**